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THE LEDGER OF THE TURNING OF THE ELEMENTS.

By Finch, of Onais.

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Communicated to the Silver Society, C.R. 981.

Severin's Twelfth Question asks: "When the four elements are turned one into another, what is it that does not change?" Every turning, he said, has its loss, and we do not know where the loss goes. For more than a hundred and fifty years the workshops of the cities have kept accounts of turnings without number, and not one of them agrees: some turnings seem to lose mana, and some workshops say they have made furnaces that bring forth mana of themselves. At Onais I have kept the accounts of turnings for thirty years: for the first eight in the old manner, and they did not agree either; afterwards I built an array chamber, and in twenty-two years made in it three hundred and eighty-six turnings, and kept the whole account of every one. I shall show three things in this paper.

First, the accounts do not agree for two reasons: in a turning some part of the mana disperses into what lies about, and has never been gathered back and measured; and those who keep the accounts take the reading of the mana meter for the whole, whereas the meter reads the density at one place, not how much was exchanged in a turning. Secondly, if in a sealed array chamber the dispersed mana is gathered back, if all mana to be entered is received into crystals of one kind and measured by discharging them through one apparatus, and if the gathering ward's own use and its leakage are measured first, then in all three hundred and eighty-six turnings the receipts and the payments agree within the error of measurement. Thirdly, the mana that the "ever-burning furnace" of Parin gives over and above its fuel comes from the crystal of its lining; when the lining is first discharged, the excess is gone.

I. WHY THE ACCOUNTS DO NOT AGREE

At first I kept the account in the old manner of the workshops: before the turning I measured the elements and the storing crystals that drive it; after it, the element obtained; and what had dispersed about I estimated from a mana meter hung in the middle of the room, read once before the turning and once after: the divisions it had risen were multiplied by the size of the room, and turned into a sum of exchange by the custom of the workshops. In this manner I kept the accounts for eight years, a hundred and six turnings, and the receipts always exceeded the payments, by some six in a hundred on the mean, and the excess was now great and now small.

Afterwards I understood that the meter cannot be so used. Fanshawe's meter reads the density of the free mana where its needle hangs. The dispersed mana does not stay evenly in the room: some runs along the walls, some passes through them, some after a while spreads elsewhere. What runs along the walls or passes through them the needle never reads, and it was left out of the account, so that the receipts were always the greater; and how dense the mana is at that moment and at that place depends on how fast it spreads, so that the excess was now great and now small. That the density has risen by so many divisions tells how dense it is at that moment and at that place, not how much this turning has dispersed in all. What is to be entered is the whole amount exchanged in a turning,

and for that all the dispersed mana must be gathered back, and received into something that can be discharged and measured.

II. THE ARRAY CHAMBER

In 958 I built an array chamber in the north of the city of Onais, twelve feet square and ten feet high, with wards for gathering dispersed mana laid upon the four walls, the ceiling and the floor. The wards lead the free mana dispersed in the chamber into the troughs of crystal upon the walls. The mana meter within hangs upon a stand that can be moved, and at every turning it is read at nine places of the room in order. Before a turning the free mana in the room is first gathered down until all nine places stand at a fixed reading, differing by no more than half a division, and after it is gathered back to the same readings, so that the room holds as much free mana before as after. The meter without hangs outside the door, and is first read for half an hour to fix its background; if during a turning it rises a division or more above the background, the ward is broken and mana has dispersed out of the room, and that turning is void, whether its account agrees or not. In twenty-two years seventeen turnings were made void, and they are not among the three hundred and eighty-six. While a turning is made, no casting is allowed within thirty paces of the chamber, lest the readings be drawn aside.

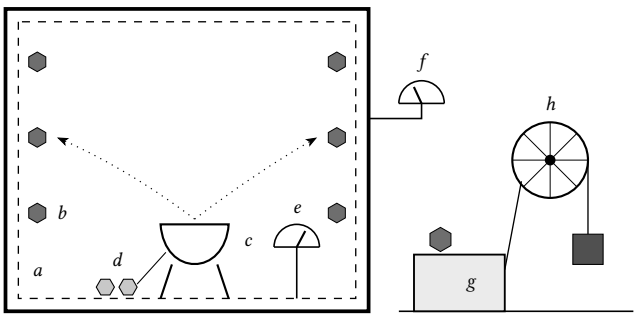
Whether the body of the chamber, the vessels and the members of the ward store up mana, I also tried: with no turning, a fixed amount of mana was let go into the room and gathered back ten times over, and what was gathered back did not grow or diminish with the number of times; and after every turning fresh crystals are set in the troughs, and the ward, fed now by a feeding crystal, goes on gathering for two hours more, and they receive nothing. The difference between what the body of the chamber, the vessels and the members of the ward hold before and after is therefore within the error of measurement, and is entered as nought.

All mana to be entered is received into crystals of one specification, not yet spent: from one vein, and of one cut, each holding some eight units when full. To measure it, the crystal is joined to a discharging apparatus and discharged entirely, and the whole amount discharged is entered; what is more than one crystal will hold is divided among several, each discharged in its turn, and the readings added. A crystal so discharged can hold no mana again, and is cast away, so that every measurement uses up several crystals. A new crystal holds a little mana of its own, and this must be taken off: from every batch ten crystals are first discharged without receiving anything, and the mean of what they give is what each holds of itself; the ten differ among themselves by no more than 0.02 unit.

The apparatus drives a lifting-wheel, which raises a standard weight; the whole height raised is the amount discharged. The apparatus has its own waste, and I have tried that it is in proportion to what is discharged: one portion of mana discharged from one crystal, and the same portion divided between two and discharged one after the other, give readings that differ within the error; discharging fast or slow gives the same reading; and each starting and stopping wastes besides some 0.01 unit, far below the error of a reading. Every portion of mana being discharged through the same apparatus, the waste is the same proportion of each, and in comparing receipts with payments it cancels.

For the unit I take a set of standard crystals kept by the Silver Society: twelve from one vein, filled at a fixed warmth of the room and by a fixed manner of receiving, and

discharged through this apparatus down to a fixed floor; the mean of what the twelve give I fix as ten units of exchange. The apparatus stops of itself at this floor, and below it a little still remains, so that a standard crystal is never spent, and after use is filled again in the fixed manner. Every month I check the apparatus again with three of them, discharging each once; and the mean of each crystal's readings in the first ten years differs from the mean in the last ten by no more than four parts in a thousand. This is the steadiness of the apparatus and the crystals together: were both to change alike, I could not tell it, of which more in Section VII. The error of one discharge is about a tenth of a unit.



a, gathering ward; *b*, troughs of crystal; *c*, vessel of the turning; *d*, storing crystals; *e*, meter within; *f*, meter without; *g*, discharging apparatus; *h*, lifting-wheel and standard weight. The chamber is twelve feet square and ten feet high.

Fig. 1. The array chamber and the instruments for measuring mana (schematic)

III. THE BLANK TRIALS

Before the turnings proper I made blank trials: in the chamber no turning was made, and only the gathering ward was kept at work, to see how much it uses of itself and how much it fails to gather.

First, its own use. The working of the ward itself takes mana, which in ordinary use it draws from the mana gathered into the troughs upon the walls, so that what is stored in the troughs is already less this portion. What the ward uses is spent upon its own working, and does not return into the room: with no mana let go, and the ward fed by a feeding crystal of its own, six hours of ordinary working put nothing into the troughs. In the blank trials the ward was fed in this way, and what the feeding crystal gave was measured: in forty trials, on the mean 0.42 unit an hour, the rate of the several trials differing by less than 0.03; and while twenty, forty or sixty units were let go into the room and the ward was busy gathering them, the rate still lay between 0.40 and 0.44. The error of the mean of the forty is some 0.005 unit an hour.

Secondly, its leakage. Ten crystals of one batch were filled in the fixed manner; five were discharged and measured, to fix what each holds, the five differing among themselves by no more than the error of reading; the mana held in the other five was released into the chamber until they would give no more, with no turning, and gathered back by the ward and measured; joined afterwards to the apparatus, the five gave nothing more. The ward was then still fed by its own crystal, and drew nothing from the troughs, so that what was gathered back contains none of its own use. Of 40.00 units let go, 39.28 were gathered back on the mean, a leakage of 1.8 in a hundred; in eight trials, letting go from twenty to sixty units, in the middle of the room, in its corners and near the ceiling, the

mean of the eight was likewise 1.8 in a hundred, and no one of them strayed from it by more than the error of reading. The portion lost is scattered out of doors, far less than a division, and the meter without cannot read it.

Both of these must be entered among the payments: the ward's own use is taken out of the mana gathered back, and the leakage is never gathered back at all; if they are not entered, the receipts will seem the greater. In the hundred and six turnings of the first eight years the dispersed mana was estimated from the meter, and these items cannot be made up; those turnings are therefore not entered in the table below, and are shown in Fig. 2 for comparison only.

IV. THE RULES OF THE LEDGER

From then on, every turning was entered by the following rules.

Receipts: the mana contained in the elements before the turning; the mana given by the storing crystals that drive the turning. The mana contained in an element is measured by taking three other portions of the same batch and the same weight, turning them back into free mana in the chamber by a working for the dispersing of elements, and gathering it by the ward; the mean of the three is taken, and they seldom differ by more than half a unit. This I call the method of dispersal. The working of dispersal is itself driven by storing crystals, and what drives it disperses into the room likewise and is gathered with the rest, so that what these crystals give must be taken off; and the residue, dispersed a second time, yields nothing, which shows that the dispersal was complete. What the storing crystals give is found as a difference: what they held before the turning is fixed by five filled crystals of the same batch, and what remains after it is discharged and measured. The turnings are all driven by storing crystals, and never by the caster's own mana, so that the caster shall not himself be entered in the account.

Payments: the mana contained in the element obtained after the turning, also measured by the method of dispersal; the dispersed mana gathered back into the troughs; the ward's own use in this turning, reckoned at 0.42 unit an hour; the leakage: what is gathered back, together with the ward's own use, is ninety-eight and two-tenths in a hundred of what was let go, and the leakage is that sum multiplied by 1.8 and divided by 98.2. What is gathered in the method of dispersal is made up for its own use and leakage in the same way, and only then is what the driving crystals gave taken off.

Since the free mana in the room is the same before and after, and the difference of what the body of the chamber, the vessels and the members of the ward hold is entered as nought, the item of what lies about is nought.

Here is the account of one turning of water into air, in units of exchange.

Receipts	*Units*	*Payments*	*Units*
Water, before the turning (by dispersal)	23.41	Air, after the turning (by dispersal)	19.87
Given by the storing crystals	8.62	Dispersed, gathered back into the troughs	10.94
		Ward's own use (two hours)	0.84
		Leakage (entered at 1.8%)	0.22
Sum	32.03	Sum	31.87

The two sides differ by 0.16 unit. Five things were measured in this turning: the portions of water, what the storing crystals held before by their fellows of the batch and what remained after, the portions of air, and the dispersed mana gathered back. Each has an error of about a tenth of a unit, greater where several crystals must be filled; this is the common scatter of many repeated measurements, not a bound. The five being independent, together they come to about a quarter of a unit. The ward's own use and the leakage, reckoned by the proportions of the blank trials, bring an error of less than 0.03. 0.16 lies within it.

V. THREE HUNDRED AND EIGHTY-SIX TURNINGS

From 958, when the chamber was finished and the blank trials made, to 980, I made by these rules three hundred and eighty-six turnings. Their workings, vessels and quantities were various, some two hundred and forty manners in all; by their elements they fall into twelve kinds, set out in Table I. The four elements are turned each into each, in both directions. All the turnings were of the ninth ring or lower.

Table I. The receipts and payments of 386 turnings (units of exchange)

Turning	*Number*	*Mean receipts*	*Mean of receipts less payments*	*Spread of the difference*
Earth into air	31	36.2	+0.04	0.27
Earth into water	28	33.9	−0.02	0.25
Earth into fire	34	41.7	+0.06	0.31
Air into earth	30	29.4	−0.03	0.24
Air into water	33	22.8	+0.01	0.21
Air into fire	37	30.5	+0.03	0.26
Water into earth	29	31.0	−0.05	0.23
Water into air	36	31.6	+0.02	0.22
Water into fire	31	35.3	+0.04	0.28
Fire into earth	32	38.8	−0.01	0.29
Fire into air	35	27.1	+0.02	0.20
Fire into water	30	34.6	−0.04	0.26

The "spread of the difference" is how far the several differences lie from their mean, taken as the root of their mean square. In all twelve kinds the mean difference lies about nought, the greatest less than a tenth of a unit; the spread lies between 0.20 and 0.31, near the quarter of a unit estimated above. Of the three hundred and eighty-six turnings, a hundred and ninety-four differ on the side of the receipts, a hundred and eighty-five on the side of the payments, and seven not at all; only three differ by more than three-quarters of a unit, where such a spread should give some two, and all three are entered as they were found. That is to say, within the error estimated, no kind of turning shows a constant gain or loss: there is none that always gives over, and none that always falls short.

Fig. 2 shows for each year from 950 to 980 the mean of receipts less payments, in hundredths of the receipts, that is, the sum of that year's differences divided by the sum of its receipts: before 958, when the dispersed mana was estimated from the meter, the receipts exceed by about six in a hundred on the mean, now more and now less; from 958, under the new rules, the mean lies about nought.

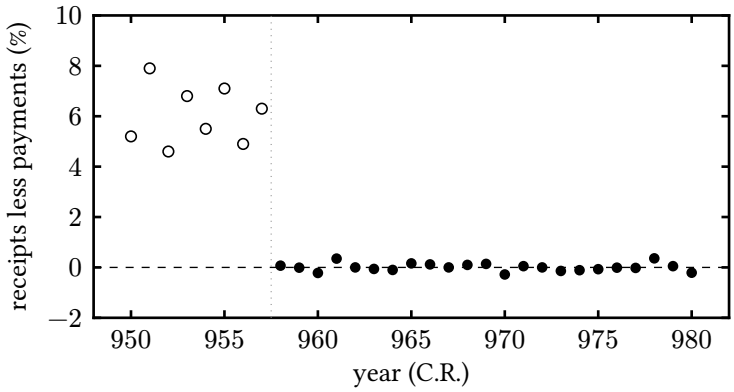


FIG. 2. Receipts less payments in each year from 950 to 980, in hundredths of that year's receipts: before 958 (open), the dispersed mana estimated from the meter; from 958 (filled), gathered back in the array chamber and entered by the rules of Section IV

VI. THE EVER-BURNING FURNACE OF PARIN

In 978 a workshop of alchemy at Parin made a furnace of which it said that, given one portion of fuel, it burned longer than that fuel should burn, and gave more mana than the fuel held; they called it the ever-burning furnace. Were this so, there would be nothing unchanging for the Twelfth Question to find. The workshop lent me one, and I kept its account in the chamber as usual.

At a fixed strength of fire the furnace burned one portion of fuel for forty hours before it went out, and gave 186.4 units; this is what was gathered back into the troughs, made up for the ward's own use in forty hours, 16.8 units, and for the leakage. The fuel, by the method of dispersal, held 151.8 units, so that 34.6 units were given over. The workshop said that the excess was what the furnace brought forth. But the lining of the hearth is a layer of crystal of the elements. The workshop had made four linings in one batch, one of them in the furnace; I discharged the other three and measured them: they held 35.6, 35.9 and 36.2 units, on the mean 35.9; after the burning, the lining in the furnace held only 1.2. The lining had lost 34.7 units, which agrees with the 34.6 given over: what the furnace gave had to be divided among some twenty crystals to be measured, and made up for forty hours of the ward's own use, the fuel too had to be divided among many crystals by the method of dispersal, and the lining could be measured only by its fellows of the batch, so that each of the two numbers has an error of about half a unit, and their difference of 0.1 lies well within it.

I then asked the workshop to put in a lining discharged beforehand, and to burn the same portion of fuel at the same strength of fire. This time the furnace went out at thirty-two hours and a half, and gave 151.5 units, 0.3 less than the fuel, within the error of this measurement, which is about half a unit, and not the least part over. The excess, then, was given out by the lining: while the furnace burned, the lining gave out its store little by little and fed the fire together with the fuel, so that the same portion of fuel burned seven hours and a half the longer. When the lining had come down to 1.2 units the furnace went out; that remainder gives itself out too slowly, I suppose, to feed the same strength of fire, but I measured only before and after the burning, and did not see whether this is what

put the fire out. The 34.6 units had been stored in the lining; the furnace did not bring them forth.

VII. CONCLUSION

From the sections above I conclude: in the turnings that I have tried, if the elements, the media that store mana, what lies about and the dispersal are entered together, the ledger of mana before and after the turning agrees within the error of measurement. The loss does not vanish; it disperses into what lies about, and can be gathered back and measured.

From this one thing more may be said, gathered from the turnings that I have tried and from this one furnace, and not proved: no instrument can bring forth mana out of nothing. A furnace, a crystal, an array can only move mana from one place to another and turn it from one form into another; whenever it is said of an instrument that it gives more than it receives, one should look first for a store that has not been entered, or for a supply from without that has not been entered.

There are some things I could not do. First, why the ledger always agrees I cannot say; I know only that in three hundred and eighty-six turnings there has not been one exception. Secondly, turnings of the tenth ring and higher are too dangerous, and I have not tried them; nor do I enter the caster's own mana, so that how the elemental pool of a caster gives and takes in a turning this paper does not treat. Thirdly, the unit of exchange hangs upon the Society's set of standard crystals and upon this one apparatus; should both one day change alike, the unit changes with them, and I could not tell it. I ask the Society to seal the standard crystals, and once a year to compare them with a few copies.